

Zach Taylor

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SUMMARY

Creative engineer with strong interests in **low-level systems**, **AI**, and application development in **C++** and **Python**. Eager to learn, grow, and expand my expertise while contributing to innovative projects that give value to customers.

SKILLS

C++, Python, Typescript, Next.js, TensorFlow, CUDA, GitHub, Linux, Jenkins, AWS, Docker, FPGA

EDUCATION

B.S. Computer Science - Oregon State University

September 2019 – June 2023

Winner of the Pioneering Concept Award at the 2023 OSU Engineering Expo

EXPERIENCE

Software Engineer Intern *Rohde & Schwarz*

April 2024 – Jan 2025

- Created **C++** network analysis solutions to address customers' limited visibility into network performance, enabling them to gather more valuable diagnostics and improve infrastructure quality.
- Designed a composite **API** to replace legacy code, achieving a **3x–4x** speed increase in benchmark tests while improving efficiency and maintainability.
- Deployed **unit tests** to identify and resolve **15** hidden computation errors affecting all software applications, maintaining software quality and preventing thousands in potential damages.
- Automated **SSH** and Ethernet configuration with **Python** scripts to eliminate repetitive manual setup, reducing development and testing time by **8%** across multiple sprints.

Undergraduate Teaching Assistant *Oregon State University*

September 2020 – June 2023

- Taught data structures and operating systems in **C** along with full stack web development in **JavaScript**, providing students with practical programming skills applicable to industry settings.
- Delivered personalized one-on-one and group tutoring to address students' struggles with core computer science concepts, improving performance by **30%** among those at risk of failing.
- Proctored exams of **100–150** students to prevent misconduct, upholding academic integrity and maintaining a fair, distraction-free testing environment.

Software Development Intern *Portland State University*

June 2019 – September 2019

- Designed a **TensorFlow** neural network model using **Python** and **OpenCV** to improve food processing efficiency and automate quality control on assembly lines.
- Optimized the model with a transfer learning approach, improving detection accuracy to **97%** in measured production tests.
- Coordinated weekly meetings with engineers and end users to ensure alignment between product development and operational goals, keeping the project on track with deadlines.

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, AI*

[GitHub](#)

- Developed an **AI** tracking robot to reduce fall incidents among elderly patients in care facilities, improving patient safety through proactive monitoring.
- Enhanced fall detection accuracy by **56%** by conducting live training in nursing homes and implementing custom **MediaPipe** functions for figure tracking.
- Mentored a team of **7–8** engineers in AI deployment and performance optimization, strengthening collaboration and technical output.
- Presented project results to Intel sponsors and showcased progress at the 2023 OSU Engineering Expo.

SIMD Image Filter | *C++, SIMD*

[GitHub](#)

- Built a **C++** image filter library to learn **SIMD** architecture and apply it to image processing.
- Implemented blur, sharpen, and edge detection algorithms to demonstrate the benefits of SIMD, achieving **5x–7x** faster performance compared to scalar versions.